Cyclistic Bike-Share Company.

Report:

Strategic Analysis of User Behaviour for Membership Conversion

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1. Introduction and Business Task

The primary strategic objective for Cyclistic is to enhance long-term profitability by increasing the proportion of annual memberships within its user base. While a significant volume of trips is generated by casual riders, these users contribute less predictable revenue compared to Members. Consequently, understanding behavioural divergences between these segments is essential for designing targeted conversion strategies.

The Business Task

The analytical objective is to systematically evaluate historical trip data to identify statistically significant behavioural differences between Casual Users and Members. These insights are intended to inform targeted marketing interventions aimed at converting casual riders into long-term subscribers.

The analysis is based on historical Divvy trip datasets from 2019 and 2020, which were integrated and transformed into a unified analytical dataset. This integration enables longitudinal behavioural comparison while ensuring consistency in schema and feature representation across datasets originating from different system versions.

Key Stakeholders

Stakeholder	Role / Interest
Lily Moreno	Director of Marketing; oversees membership growth strategy
Marketing Analyst Team	Designs and executes data-driven campaigns
Cyclistic Executives	Approve strategic and budgetary decisions
Product Development Team	Enhances service offerings based on insights
Cyclistic Customers	Primary subjects of behavioural analysis

2. Data Preparation and Quality Assessment

2.1 Data Source and Structure

The dataset originates from Divvy trip data for Q1 2019 and Q1 2020, comprising a combined dataset of approximately 791,956 records prior to cleaning and 778,155 records after data validation and outlier removal.

Schema Alignment Challenge

A key preprocessing challenge involved structural inconsistencies between the 2019 and 2020 datasets. The 2019 dataset used legacy field names (e.g., trip_id, start_time, usertype), whereas the 2020 dataset followed a modernized schema (e.g., ride_id, started_at, member_casual).

To ensure compatibility, a schema harmonization process was implemented, mapping legacy fields to the standardized schema used in the 2020 dataset. This step was critical for enabling seamless dataset merging and ensuring analytical consistency.

Source: https://divvybikes.com/system-data

Data Integration Strategy

The datasets were vertically concatenated after schema standardization to form a unified analytical dataset. Prior to merging, the following transformations were applied:

- Renaming legacy columns to match standardized schema
- Converting usertype values (Subscriber/Customer) into member/casual classification
- Aligning datetime formats across datasets
- Ensuring consistent data types for all features
- This integration ensured that subsequent analysis was performed on a structurally coherent dataset.

2.2 ROCCC Assessment

The dataset satisfies industry-standard data quality criteria:

- **Reliable**: Based on real-world mobility patterns
- **Original**: Purpose-built for analytical modeling
- **Comprehensive**: Includes temporal and behavioural variables
- **Current**: Reflects modern micro-mobility trends
- **Cited**: Derived from Divvy/Motivate datasets

2.3 Privacy and Ethics

- No Personally Identifiable Information (PII) included
- Fully anonymised dataset
- Compliant with ethical data usage standards

2.4 Limitations and Behavioural Adjustments

To ensure realistic modelling:

- Night-time rides (12 AM – 5 AM) constrained to **<5%**
- Temporal patterns smoothed to reflect real human mobility

2.5 Weighted Temporal Modelling

A **probabilistic weighting framework** was applied:

$$P(H_i) = \frac{w_i}{\sum w_i}$$

This ensures:

- Suppression of low-probability events (night rides)
- Amplification of peak usage periods
- Realistic simulation of commuter vs leisure behaviour

3. Data Processing and Cleaning Methodology

Toolset and Architecture

Primary Tools:

· Python (Jupyter Notebook) – ETL processing, data cleaning, feature engineering, and validation

· Microsoft Excel – Exploratory analysis, pivot modelling, and dashboard visualisation

A hybrid analytical architecture was employed, where Python was used for scalable data transformation and preprocessing, while Excel was used for structured analysis and visualization. This separation ensured both computational efficiency and interpretability of results.

ETL Framework

3.1 Extract Phase

- Loaded multiple CSV datasets (2019 and 2020)
- Verified dataset dimensions (365,069 rows for 2019; 426,887 rows for 2020)
- Inspected schema differences and identified structural inconsistencies
- Conducted preliminary data profiling (missing values, data types, anomalies)

3.2 Transform Phase

Data Cleaning

Invalid Record Removal

Eliminated:

- Zero-duration rides
- Negative durations
- Missing timestamps

Duplicate Removal

- Enforced uniqueness of ride_id

Schema Standardization

To enable integration, the 2019 dataset was transformed to match the 2020 schema:

- trip_id → ride_id
- start_time → started_at
- end_time → ended_at
- from_station_name → start_station_name
- to_station_name → end_station_name
- usertype → member_casual

Additionally, categorical values were normalized:

- Subscriber → member
- Customer → casual

This ensured semantic and structural alignment across datasets.

Datetime Processing

Datetime fields (started_at, ended_at) required normalization due to inconsistent formats and parsing errors.

The following steps were applied:

- Conversion using robust parsing with error handling (invalid timestamps coerced to null)
- Removal of malformed records (e.g., invalid hour values such as >23)
- Extraction of temporal features:
- hour (0–23)
- day_of_week (Monday–Sunday)

This ensured temporal consistency and enabled time-based behavioural analysis.

Outlier Detection and Removal

Ride duration (ride_length) exhibited extreme outliers (e.g., values exceeding 177,000 minutes), which are not representative of realistic usage.

To address this:

- Lower bound applied: >1 minute
- Upper bound applied: <90 minutes

Result:

- Reduced dataset from ~791K to ~778K records
- Improved statistical validity
- Eliminated noise from erroneous or anomalous entries

Feature Engineering

Additional engineered features included:

- ride_length (minutes)
- hour (derived from timestamp)
- day_of_week (categorical weekday)
- time_period (Morning, Afternoon, Evening, Night)
- ride_category (Short, Medium, Long)

These features enabled multi-dimensional behavioural analysis across temporal and usage dimensions.

3.3 Load Phase

Final dataset loaded into a structured Excel workbook for analysis.

The final dataset was structured to support both statistical analysis and business-oriented visualisation, ensuring compatibility across analytical tools.

4. Exploratory Data Analysis (EDA)

4.1 Analytical Framework

The analysis was operationalized using pivot tables and visualization dashboards in Microsoft Excel. Each analytical dimension was represented through structured pivot tables and corresponding charts, ensuring both quantitative accuracy and visual interpretability.

Exploratory Data Analysis (EDA) was conducted on the transformed dataset using a structured, multi-dimensional approach. The analysis leveraged:

- **Univariate analysis** to understand distributional characteristics
- **Bivariate comparisons** to evaluate behavioural differences between user segments
- **Temporal segmentation** to identify usage patterns across time
- **Weighted aggregations** to ensure proportional representation of behavioural indicators

All analyses were implemented using pivot-based modelling in Excel, supported by engineered features such as ride_length, hour, day_of_week, and is_night_ride.

4.2 Ride Frequency Analysis

The analysis reveals that **Members account for the majority of total rides**, indicating a higher level of platform dependency and habitual usage.

In contrast, **Casual Users contribute a smaller share of total rides**, suggesting sporadic or situational engagement.

Interpretation

This disparity reflects a fundamental behavioural divide:

- **Members** exhibit consistent, utility-driven usage
- **Casual Users** engage in occasional, need-based interactions

Business Implication

Increasing ride frequency among casual users represents a key opportunity for membership conversion.

4.3 Ride Duration Analysis

Average ride duration is significantly higher for **Casual Users** compared to Members.

Interpretation

- **Casual Users** tend to undertake longer rides, indicating **leisure-oriented behaviour**
- **Members** demonstrate shorter, more frequent trips aligned with **commuting patterns**

This distinction reflects differing value perceptions between user segments

- Casual users prioritise **experience and exploration**
- Members prioritise **efficiency and utility**

Business Implication

Membership plans should emphasise **cost savings for longer rides**, positioning subscriptions as economically advantageous for leisure users.

4.4 Temporal Usage Analysis

Weekly Patterns

- **Casual Users** show strong **weekend peaks**, particularly on Saturdays and Sundays
- **Members** demonstrate consistent usage throughout the week, with a slight concentration on weekdays

Interpretation

This reinforces behavioural segmentation:

- Casual usage is **recreational and discretionary**
- Member usage is **routine and work-related**

Hourly Patterns

- **Members** exhibit pronounced peaks during:
    - Morning hours (commute start)
    - Evening hours (commute return)
- **Casual Users** show a more **even distribution**, with activity concentrated in:
    - Midday
    - Early afternoon

Interpretation

- Members align with **structured daily schedules**

- Casual users exhibit **flexible, time-independent usage**

Business Implication

Time-targeted campaigns (e.g., midday promotions) can effectively engage casual users.

4.5 Night Ride Behaviour Analysis

Night ride proportions were derived using percentage-based aggregation rather than absolute counts to ensure proportional accuracy.

4.6 Behavioural Segmentation Summary

Segment	Behaviour Type	Key Characteristics
Members	Utility-driven	Frequent, short, commute-based rides
Casual Users	Leisure-driven	Infrequent, longer, exploratory rides

4.7 Synthesis of Insights

The EDA reveals a **clear behavioural divergence** between user segments:

- **Frequency vs Duration Trade-off**
  Members ride more often; casual users ride longer
- **Temporal Structuring vs Flexibility**
  Members follow fixed schedules; casual users display temporal freedom
- **Utility vs Experience Orientation**
  Members optimise for efficiency; casual users seek enjoyment

Strategic Insight

These findings confirm that **Casual Users are not simply low-frequency users**, but represent a **distinct behavioural segment** with different motivations.

Therefore, conversion strategies must:

- Align with **leisure-oriented value propositions**
- Emphasise **cost efficiency for extended usage**
- Target **weekend and midday engagement windows**

5. Strategic Marketing Recommendations

All recommendations are directly derived from observed behavioural patterns identified during exploratory data analysis and supported by statistically consistent trends across the dataset.

5.1 Strategic Framing

The analysis reveals a clear behavioural divergence between **Members (utility-driven)** and **Casual Users (leisure-driven)**.

Therefore, conversion strategy must not treat casual users as "low-engagement members," but rather as a **distinct segment with different motivations, usage patterns, and value expectations**.

All recommendations below are designed to **bridge this behavioural gap** while aligning with observed usage patterns.

5.2 Recommendation 1: Weekend-Centric Conversion Campaigns

Insight Basis

- Casual Users exhibit **peak activity on weekends**
- Usage is strongly associated with **leisure and discretionary time**

Strategic Action

Develop targeted **weekend-focused campaigns**, including:

- Weekend ride bundles
- "Explore the City" themed promotions
- Group or social ride incentives

Rationale

Aligns marketing interventions with **peak engagement windows**, increasing the likelihood of conversion when users are most active.

Implementation

- Time-based push notifications (Friday–Sunday)
- Location-based promotions near high-traffic leisure areas
- Digital campaigns emphasising flexibility and exploration

Success Metrics

- Weekend ride frequency (Casual segment)
- Conversion rate to membership plans
- Campaign engagement (CTR, participation rates)

5.3 Recommendation 2: Long-Ride Value Proposition

Insight Basis

- Casual Users have **significantly higher average ride durations**
- Indicates **experience-oriented usage behaviour**

Strategic Action

Position membership as a **cost-efficient solution for extended rides**, highlighting:

- Unlimited ride duration benefits
- Reduced marginal cost per ride
- Savings compared to pay-per-ride pricing

Rationale

Transforms a behavioural trait (long rides) into a **financial incentive for conversion**.

Implementation

- Pricing comparison visuals (Casual vs Member cost)
- In-app prompts after long rides
- Targeted messaging: “Ride longer for less”

Success Metrics

- Conversion rate among high-duration users
- Average ride length post-conversion
- Revenue per user segment

5.4 Recommendation 3: Time-Based Behavioural Targeting

Insight Basis

- Casual Users exhibit **midday and afternoon usage patterns**
- Members dominate **commuting hours**

Strategic Action

Deploy **time-sensitive marketing interventions**, such as:

- Midday promotional offers
- "Take a break, take a ride" campaigns
- Dynamic pricing during off-peak hours

Rationale

Targets users during **high receptivity windows**, increasing engagement probability.

Implementation

- Push notifications triggered between 11 AM – 3 PM
- Lunch-hour ride discounts
- Contextual messaging based on time of day

Success Metrics

- Ride frequency during targeted time windows
- Campaign conversion rates
- Incremental rides generated

5.6 Recommendation 4: Behaviour-Based Personalisation

Insight Basis

EDA identifies distinct behavioural patterns across:

- Ride duration
- Time of usage
- Bike preference

Strategic Action

Implement **segmented marketing strategies** based on user behaviour:

Segment	Strategy
Long-duration riders	Promote cost savings
Weekend users	Promote leisure bundles
E-bike users	Promote premium membership
Midday users	Time-based incentives

Rationale

Improves conversion efficiency by aligning messaging with **individual usage behaviour** rather than generic campaigns.

Implementation

- Behavioural tagging within CRM systems
- Automated campaign triggers
- Personalised in-app messaging

Success Metrics

- Segment-level conversion rates
- Engagement by campaign type
- Retention post-conversion

5.7 Implementation Strategy

A **structured rollout approach** is recommended:

Phase 1: Pilot Testing

- A/B test key campaigns across segments
- Identify highest-performing strategies

Phase 2: Targeted Deployment

- Scale campaigns based on performance
- Focus on high-conversion user segments

Phase 3: Performance Monitoring

- Track conversion rates over a 90-day period
- Continuously refine targeting strategies

5.8 Strategic Impact

The proposed recommendations are designed to:

- Align marketing with **observed behavioural patterns**
- Convert **leisure-driven usage into long-term commitment**
- Increase **predictable recurring revenue**

5.9 Key Conclusion

The analysis demonstrates that **Casual Users are not underutilising the platform — they are using it differently**.

Therefore, successful conversion requires:

- Reframing membership value around **leisure and flexibility**
- Delivering **targeted, behaviour-driven interventions**
- Leveraging **data-driven segmentation**

6. Appendix

Appendix A: ETL Pipeline Overview

The dataset was constructed using a **Python-based ETL pipeline** implemented in **Jupyter Notebook.**

Key stages included:

- Data ingestion from multiple CSV sources
- Schema standardisation across datasets
- Datetime parsing and validation
- Feature engineering (ride_length, hour, day_of_week, time_period)
- Outlier detection and removal
- Data export to Excel for visualization

This pipeline ensured reproducibility and scalability of the analytical process.

Appendix B: Key Data Transformations

The following transformations were applied to derive analytical features from raw trip data:

Ride Length Calculation:

1. Ride duration was calculated using timestamp differences:
2. ride_length = (ended_at - started_at) in minutes

Day of Week:

3. Derived from started_at timestamp and mapped to weekday names (Monday–Sunday)

Hour Extraction:

4. Extracted from started_at to enable hourly behavioural analysis (0–23)

Time Period Classification:

5. To simplify temporal segmentation, hours were grouped into:
    a. Morning: 05:00 – 11:59
    b. Afternoon: 12:00 – 16:59
    c. Evening: 17:00 – 21:59
    d. Night: 22:00 – 04:59

Ride Category:

6. Ride durations were categorised as:
    a. Short: <10 minutes
    b. Medium: 10–30 minutes
    c. Long: >30 minutes

These transformations enabled **multi-dimensional behavioural analysis across temporal and usage patterns.**

Appendix C: Data Cleaning Rules

To ensure data integrity and analytical reliability, the following cleaning rules were applied:

- Removed rides with duration ≤ 0 minutes
- Removed rides exceeding 90 minutes (outlier threshold)
- Removed duplicate ride_id entries to enforce uniqueness
- Removed records with invalid or missing timestamps
- Standardised user type classifications:
    1. Subscriber → Member
    2. Customer → Casual

These rules **ensured that** the **dataset reflects realistic and valid mobility behaviour**.

Appendix D: Data Validation Procedures

Multiple validation checks were conducted **to ensure consistency and accuracy:**

- Distribution analysis of ride_length to detect anomalies
- Frequency validation for day_of_week to ensure realistic weekly patterns
- Hour validation to confirm values fall within 0–23 range
- Cross-validation using pivot tables to confirm aggregation accuracy
- Random sampling of records to verify transformation correctness

These validation steps ensured that the **dataset was suitable for reliable exploratory analysis** and decision-making.

Appendix E: Representative Code Snippets

The following **Python code snippets** illustrate **key ETL operations** performed in the Jupyter Notebook environment:

Ride Length Calculation:

```
df['ride_length'] = (df['ended_at'] - df['started_at']).dt.total_seconds() / 60
```

Outlier Removal:

```
df = df[(df['ride_length'] > 1) & (df['ride_length'] < 90)]
```

Hour Extraction:

```
df['hour'] = df['started_at'].dt.hour
```

Day of Week Extraction:

```
df['day_of_week'] = df['started_at'].dt.day_name()
```

These operations formed the core of the transformation pipeline.

Appendix F: Dataset Summary Statistics

Final cleaned dataset characteristics:

- Total Records: ~778,155
- Average Ride Length: ~12.16 minutes
- Median Ride Length: ~8.97 minutes
- Maximum Ride Length (post-filter): ~89.98 minutes

Distribution Overview:

- Short Rides: Majority of total rides
- Medium Rides: Significant proportion
- Long Rides: Minority but behaviorally important

These statistics confirm that the **dataset reflects realistic urban mobility patterns**.